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CSPB 4502: Data Mining

Project Part 1: Project Proposal

Discovering Fitness Profiles and Workout Performance Patterns Through Data Mining

Description

This project analyzes the Gym Members Exercise Dataset to identify patterns in fitness behavior and workout performance. The study will investigate whether gym members can be grouped into meaningful fitness profiles based on workout habits and physical characteristics, determine which factors most strongly influence calories burned during exercise, and evaluate whether a member's experience level can be predicted from fitness and workout-related attributes. The results may provide insights into workout effectiveness and exercise behavior.

Prior Work

Data mining and machine learning techniques have been widely applied in health and fitness analytics. Researchers have used clustering methods to identify fitness profiles and predictive models to estimate exercise outcomes and energy expenditure. Bonomi et al. (2012) demonstrated how activity-monitoring data can be used to predict physical activity patterns and energy expenditure. This project builds upon similar concepts by applying clustering, classification, and regression techniques to a gym member fitness dataset.

Citation: Bonomi, A. G., Plasqui, G., Goris, A. H., & Westerterp, K. R. (2012). Improving assessment of daily energy expenditure by identifying types of physical activity with a single accelerometer. Journal of Applied Physiology, 107(3), 655-661.

Dataset

Dataset Name: Gym Members Exercise Dataset

Source: Kaggle

URL: https://www.kaggle.com/datasets/valakhorasani/gym-members-exercise-dataset

Data Supplier

The dataset is publicly available through Kaggle and contains fitness and workout information for gym members.

Dataset Size

- 973 observations (gym members)
- 15 attributes

Key attributes include:

- Age
- Gender
- Weight (kg)
- Height (m)
- BMI
- Workout Type
- Workout Frequency
- Session Duration
- Calories Burned
- Experience Level
- Body Fat Percentage
- Water Intake
- Heart Rate Metrics (Maximum, Average, and Resting BPM)

Download Status

The dataset has been downloaded to my personal computer and successfully loaded into Python using Pandas. Preliminary exploration found no missing values across any of the dataset attributes.

Proposed Work

Data Cleaning

- Verify data quality and consistency
- Check for duplicate records
- Validate numerical ranges and data types

Data Preprocessing

- Encode categorical variables such as gender and workout type
- Normalize numerical variables when appropriate
- Prepare data for machine learning models

Exploratory Data Analysis

- Generate descriptive statistics
- Analyze distributions of fitness and workout variables
- Examine relationships among attributes using correlation analysis and visualization techniques

Clustering

- Apply K-Means clustering to identify fitness profiles among gym members
- Compare clusters using BMI, workout frequency, body fat percentage, and calories burned

Classification

- Build classification models to predict a member’s experience level using fitness and workout characteristics

Regression

- Develop regression models to predict calories burned during exercise sessions

Visualization

- Correlation heatmaps
- Scatter plots
- Histograms
- Cluster visualizations
- Feature importance charts

Preliminary Findings

Preliminary exploratory analysis identified strong positive correlations between calories burned and session duration ($r = 0.91$), experience level ($r = 0.69$), and workout frequency ($r = 0.58$). A moderate negative correlation was observed between body fat percentage and calories burned ($r = -0.60$). These findings suggest the dataset contains meaningful relationships suitable for clustering, classification, and predictive modeling.

Tools

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Git and GitHub for version control

Evaluation

Clustering performance will be evaluated using silhouette scores and interpretation of resulting fitness profiles. Classification models will be evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Regression models will be evaluated using R^2 and Root Mean Squared Error (RMSE). The success of the project will be measured by the ability to discover meaningful relationships between fitness characteristics, workout habits, and exercise outcomes.